

```

#define SERVER_FIFO "server_fifo"
#define MESSAGE_SIZE 256

typedef struct {
    pid_t client_pid;
    char message[MESSAGE_SIZE];
} Request;

void handle_client(Request request) {

    char client_fifo[100];

    // Create the client's response FIFO name
    snprintf(client_fifo, sizeof(client_fifo),
             "client_%d_fifo", request.client_pid);

    // Open client's FIFO for writing
    int fd = open(client_fifo, O_WRONLY);

    if (fd == -1) {
        perror("open client FIFO");
        exit(1);
    }

    char response[512];

    snprintf(response, sizeof(response),
             "Server received message from Client %d: \"%s\"\n",
             request.client_pid,
             request.message);

    write(fd, response, strlen(response));

    close(fd);

    printf("Processed request from Client %d\n",

```

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <sys/wait.h>
#include <string.h>
#include <errno.h>

#define SERVER_FIFO "server_fifo"
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    }

    char response[512];

    snprintf(response, sizeof(response),
             "Server received message from Client %d: \"%s\"\n",

```

```

        request.client_pid,
        request.message);

write(fd, response, strlen(response));

close(fd);

printf("Processed request from Client %d\n",
        request.client_pid);

exit(0);
}

```

```

int main() {

    printf("=====\n");
    printf("        FIFO SERVER STARTED\n");
    printf("=====\n");

    // Remove old FIFO if it exists
    unlink(SERVER_FIFO);

    // Create named pipe
    if (mkfifo(SERVER_FIFO, 0666) == -1) {
        perror("mkfifo");
        return 1;
    }

    printf("Server FIFO created: %s\n", SERVER_FIFO);
    printf("Waiting for clients...\n\n");

    // Open server FIFO for reading
    int server_fd = open(SERVER_FIFO, O_RDONLY);

    if (server_fd == -1) {
        perror("open server FIFO");
        unlink(SERVER_FIFO);
    }
}

```

```
while (1) {

    Request request;

    ssize_t bytes_read = read(
        server_fd,
        &request,
        sizeof(Request)
    );

    if (bytes_read == sizeof(Request)) {

        printf("Request received from Client %d\n",
            request.client_pid);

        printf("Message: %s\n",
            request.message);

        // Create a child process to handle the client
        pid_t pid = fork();

        if (pid == -1) {
            perror("fork");
        }
        else if (pid == 0) {
            close(server_fd);

            handle_client(request);
        }
    }

    close(server_fd);
    unlink(SERVER_FIFO);

    return 0;
}
```

```
shreyas@Grigori:~$ mkdir -p ~/fifo_client_server
shreyas@Grigori:~$ cd ~/fifo_client_server
shreyas@Grigori:~/fifo_client_server$ nano server.c
shreyas@Grigori:~/fifo_client_server$ nano client.c
shreyas@Grigori:~/fifo_client_server$ gcc server.c -o server
shreyas@Grigori:~/fifo_client_server$ gcc client.c -o client
shreyas@Grigori:~/fifo_client_server$ ls
client  client.c  server  server.c
shreyas@Grigori:~/fifo_client_server$ cd ~/fifo_client_server
shreyas@Grigori:~/fifo_client_server$ ./server
=====
      FIFO SERVER STARTED
=====
Server FIFO created: server_fifo
Waiting for clients...

Request received from Client 1170
Message: Hello Server
Processed request from Client 1170
Request received from Client 1198
Message: hi
Processed request from Client 1198
^C
shreyas@Grigori:~/fifo_client_server$ |
```

```
shreyas@Grigori:~$ cd ~/fifo_client_server
shreyas@Grigori:~/fifo_client_server$ ./client
=====
                FIFO CLIENT
=====
Client PID: 1170
Enter message: Hello Server

Message sent to server.

Server Response:
Server received message from Client 1170: "Hello Server"
shreyas@Grigori:~/fifo_client_server$
```

```
shreyas@Grigori:~$ cd ~/fifo_client_server
shreyas@Grigori:~/fifo_client_server$ ./client
=====
FIFO CLIENT
=====
Client PID: 1198
Enter message: hi

Message sent to server.

Server Response:
Server received message from Client 1198: "hi"
shreyas@Grigori:~/fifo_client_server$ ls -l
total 48
-rwxr-xr-x 1 shreyas shreyas 16544 Sep 20 14:52 client
-rw-r--r-- 1 shreyas shreyas  1951 Sep 20 14:51 client.c
-rwxr-xr-x 1 shreyas shreyas 16536 Sep 20 14:52 server
-rw-r--r-- 1 shreyas shreyas  2431 Sep 20 14:51 server.c
prw-r--r-- 1 shreyas shreyas      0 Sep 20 14:54 server_fifo
shreyas@Grigori:~/fifo_client_server$
```